

$$R^2$$

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1 r^2

The square of the correlation coefficient r is typically used for simple linear regression because of the limits of its algebraic definition. It measures how close two lists of numbers are to a straight line. There does not need to be however a linear relationship between the two lists of numbers.

Let $a_i = x_i - \bar{x}$ and $b_i = y_i - \bar{y}$ for $i = 1, \dots, n$, where $\bar{x}$ and $\bar{y}$ are the sample averages. Then,

$$r = \frac{\sum_{i=1}^n a_i b_i}{\sqrt{\sum_{i=1}^n a_i^2} \sqrt{\sum_{i=1}^n b_i^2}} \quad (1)$$

$$r^2 = \frac{(\sum_{i=1}^n a_i b_i)^2}{(\sum_{i=1}^n a_i^2) (\sum_{i=1}^n b_i^2)} \quad (2)$$

But by Cauchy-Schwarz, which caps the dot product,

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right) \quad (3)$$

And so dividing both sides of the Cauchy-Schwarz expression by $(\sum a_i^2) (\sum b_i^2) > 0$ gives

$$0 \leq r^2 \leq 1 \quad \text{and therefore} \quad -1 \leq r \leq 1. \quad (4)$$

2 R^2

We can generalize r^2 for use in e.g. multiple linear regression. That is R^2 .

Notation. There are n observations. The actual outcomes are y_i , with average $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$. The model's predictions are $\hat{y}_i$. The error (residual) of observation i is $e_i = y_i - \hat{y}_i$.

Sums of squares.

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2 \quad (\text{total sum of squares}) \quad (5)$$

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (\text{residual sum of squares}) \quad (6)$$

$$\text{ESS} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 \quad (\text{explained sum of squares}) \quad (7)$$

General definition (valid for any model and any data set on which the predictions are scored):

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} \quad (8)$$

In this form, $R^2 \leq 1$. There is no lower bound.

In the general definition, R^2 indicates the share of variation explained.

Squared correlation form. Under the same conditions, R^2 equals the squared correlation between y and $\hat{y}$:

$$R^2 = r_{y\hat{y}}^2 = \frac{[\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})]^2}{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2} \quad (9)$$

where $\bar{\hat{y}}$ is the average of the predictions.

3 Applications of R^2

Most applications below show that R^2 can be below -1.

Simple linear regression. With one predictor x , an intercept and ordinary least squares, R^2 equals the squared correlation between x and y :

$$R^2 = r_{xy}^2 = \frac{S_{xy}^2}{S_{xx} S_{yy}}, \quad (10)$$

where $S_{xx} = \sum (x_i - \bar{x})^2$, $S_{yy} = \sum (y_i - \bar{y})^2$ and $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$.

The above assumes idealized OLS was used for the linear relationship. If for instance the slope is different from OLS, then R^2 can be below -1, but still can't

be above 1. It can fall below -1 because the 'fit' now has to consider both (1) how well the points are spread around the fitted line, and (2) how well the fitted line is different from the idealized OLS line.

Non-linear regression. The closer the non-linear regression fit to the actual points, the closer R^2 approaches 1. But if it fits badly, it will be negative and can fall below -1.

Out-of-sample form. For a test set of m observations y_j with predictions $\hat{y}_j$ that were not fitted to those observations:

$$R_{\text{test}}^2 = 1 - \frac{\sum_{j=1}^m (y_j - \hat{y}_j)^2}{\sum_{j=1}^m (y_j - \bar{y}_{\text{test}})^2} \quad (11)$$

Here $\bar{y}_{\text{test}}$ is the average of the test outcomes. Some authors use the average of the training outcomes in its place. This form can be negative. $R^2 = 1$ means perfect fit, $R^2 = 0$ can mean fitting the test set data, and negative means poor fit. Below -1 means very poor fit. In this case, the R^2 no longer means the share of variation explained; rather, it's about how much smaller the model's squared error is than that of the baseline.

For the sake of completeness:

Adjusted R^2 . With k predictors (not counting the intercept):

$$R_{\text{adj}}^2 = 1 - (1 - R^2) \frac{n - 1}{n - k - 1} \quad (12)$$

This value can be negative even when $R^2 \geq 0$.

4 Generalized R^2

R^2 can be below -1. This happens if $RSS > TSS$. But how? If we generalize the formula of R^2 by using the general definition of TSS.

Decomposition of TSS.

$$TSS = RSS + ESS + 2 \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) \quad (13)$$

The last term is called the cross-term.

Derivation of the cross-term in the decomposition of TSS.

Notation. There are n observations. The actual outcomes are y_i , with average

$\bar{y}$. The predictions are $\hat{y}_i$, and the errors are $e_i = y_i - \hat{y}_i$. Define

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad (14)$$

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (15)$$

$$\text{ESS} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2. \quad (16)$$

Step 1: Add and subtract the prediction. For each observation,

$$y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y}). \quad (17)$$

Step 2: Square both sides. For any two numbers u and v , $(u + v)^2 = u^2 + v^2 + 2uv$. Let $u = y_i - \hat{y}_i$ and $v = \hat{y}_i - \bar{y}$. Then

$$(y_i - \bar{y})^2 = (y_i - \hat{y}_i)^2 + (\hat{y}_i - \bar{y})^2 + 2(y_i - \hat{y}_i)(\hat{y}_i - \bar{y}). \quad (18)$$

The last term is the product uv . It appears because squaring a sum of two terms produces a mixed product of the two terms.

Step 3: Sum over all observations.

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + 2 \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}). \quad (19)$$

Step 4: Name the terms.

$$\text{TSS} = \text{RSS} + \text{ESS} + \underbrace{2 \sum_{i=1}^n e_i (\hat{y}_i - \bar{y})}_{\text{cross-term}}. \quad (20)$$

5 When the cross-term is zero

Step 5: Split the sum. Write $C = \sum_{i=1}^n e_i (\hat{y}_i - \bar{y})$. Then

$$C = \sum_{i=1}^n e_i \hat{y}_i - \bar{y} \sum_{i=1}^n e_i. \quad (21)$$

Step 6: Least squares conditions. Let the model be $\hat{y}_i = b_0 + b_1 x_{i1} + \dots + b_k x_{ik}$. The coefficients are chosen to minimise RSS. Setting the derivative of

RSS with respect to each coefficient to zero gives

$$\frac{\partial \text{RSS}}{\partial b_0} = -2 \sum_{i=1}^n e_i = 0 \quad (\text{condition A, requires the intercept } b_0), \quad (22)$$

$$\frac{\partial \text{RSS}}{\partial b_j} = -2 \sum_{i=1}^n x_{ij} e_i = 0 \quad \text{for } j = 1, \dots, k \quad (\text{condition B}). \quad (23)$$

Step 7: Show that $C = 0$. By condition A, the second part of C is $\bar{y} \sum e_i = 0$. For the first part,

$$\sum_{i=1}^n e_i \hat{y}_i = b_0 \sum_{i=1}^n e_i + b_1 \sum_{i=1}^n x_{i1} e_i + \dots + b_k \sum_{i=1}^n x_{ik} e_i = 0, \quad (24)$$

because every sum on the right is zero by conditions A and B. Therefore $C = 0$, and

$$\text{TSS} = \text{RSS} + \text{ESS}. \quad (25)$$

Conditions required. The model is linear in its coefficients, is fitted by ordinary least squares, includes an intercept, and the sums of squares are calculated on the data used for fitting. If any of these conditions fails, the cross-term can be positive or negative.

Special case. Suppose the model is linear in its coefficients, is fitted by ordinary least squares, includes an intercept, and is scored on the data used for fitting. Then the cross-term is zero, so

$$\text{TSS} = \text{RSS} + \text{ESS}, \quad (26)$$

and the following equivalent forms hold:

$$R^2 = \frac{\text{ESS}}{\text{TSS}} \quad (27)$$

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} \quad (28)$$

$$0 \leq R^2 \leq 1 \quad (29)$$

6 When the cross-term is NOT zero

Limits on the cross-term from the Cauchy–Schwarz inequality.

Notation. There are n observations. The actual outcomes are y_i , with average $\bar{y}$. The predictions are $\hat{y}_i$. Define the deviations from the average of the actual outcomes:

$$a_i = y_i - \bar{y}, \quad d_i = \hat{y}_i - \bar{y}, \quad e_i = y_i - \hat{y}_i = a_i - d_i. \quad (30)$$

Then

$$\text{TSS} = \sum_{i=1}^n a_i^2, \quad \text{ESS} = \sum_{i=1}^n d_i^2, \quad \text{RSS} = \sum_{i=1}^n (a_i - d_i)^2. \quad (31)$$

The decomposition from before is

$$\text{TSS} = \text{RSS} + \text{ESS} + 2C, \quad C = \sum_{i=1}^n e_i d_i, \quad (32)$$

and $2C$ is the cross-term.

Step 1: Write C with the vectors a and d .

Since $e_i = a_i - d_i$,

$$C = \sum_{i=1}^n (a_i - d_i) d_i = \sum_{i=1}^n a_i d_i - \text{ESS}. \quad (33)$$

Step 2: Apply the Cauchy–Schwarz inequality.

The Cauchy–Schwarz inequality states that

$$\left| \sum_{i=1}^n a_i d_i \right| \leq \sqrt{\sum_{i=1}^n a_i^2} \sqrt{\sum_{i=1}^n d_i^2} = \sqrt{\text{TSS} \cdot \text{ESS}}. \quad (34)$$

Define two quantities (assume $\text{ESS} > 0$; if $\text{ESS} = 0$, every $d_i = 0$, so $C = 0$ and $R^2 = 0$):

$$s = \sqrt{\frac{\text{ESS}}{\text{TSS}}} \geq 0 \quad (\text{length of } d \text{ divided by length of } a), \quad (35)$$

$$\rho = \frac{\sum_i a_i d_i}{\sqrt{\text{TSS} \cdot \text{ESS}}} \quad (\text{cosine of the angle between } a \text{ and } d). \quad (36)$$

Cauchy–Schwarz is the statement that $-1 \leq \rho \leq 1$. When the average of the predictions equals $\bar{y}$, ρ is the ordinary correlation between y and $\hat{y}$.

Step 3: An exact formula for C .

Since $\sum_i a_i d_i = \rho \sqrt{\text{TSS} \cdot \text{ESS}} = \rho s \text{TSS}$ and $\text{ESS} = s^2 \text{TSS}$,

$$\boxed{C = \text{TSS } s (\rho - s)} \quad (37)$$

Step 4: Bounds on the cross-term.

Using $-1 \leq \rho \leq 1$ in the exact formula:

$$-\text{TSS}(s + s^2) \leq C \leq \text{TSS}(s - s^2). \quad (38)$$

Upper bound. The function $s - s^2$ has its maximum value $1/4$ at $s = 1/2$. Therefore

$$C \leq \frac{\text{TSS}}{4}, \quad \text{so} \quad 2C \leq \frac{\text{TSS}}{2}. \quad (39)$$

Equality holds only when $\rho = 1$ and $s = 1/2$, that is, when $d_i = a_i/2$ for all i . This is the case of predictions compressed to half of the actual vertical spread.

Lower bound. The expression $-\text{TSS}(s + s^2)$ becomes arbitrarily negative as s increases. Nothing limits the cross-term from below, because nothing limits the length of the predicted deviations d .

Step 5: The sign of the cross-term.

For $s > 0$, the sign of C is the sign of $\rho - s$:

$$\begin{aligned} \rho > s & \quad C > 0 \text{ (predictions are shorter than their alignment with } a \text{ allows)} \\ \rho = s & \quad C = 0 \\ \rho < s & \quad C < 0 \text{ (predictions are longer than their alignment with } a \text{ allows)} \end{aligned}$$

Step 6: R^2 in terms of s and ρ .

From $\text{TSS} = \text{RSS} + \text{ESS} + 2C$,

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = \frac{\text{ESS} + 2C}{\text{TSS}} = s^2 + 2s(\rho - s) \quad (40)$$

$$\boxed{R^2 = 2\rho s - s^2} \quad (41)$$

This is the law of cosines applied to the triangle formed by a , d and $a - d$: $\text{RSS} = \text{TSS} + \text{ESS} - 2\sqrt{\text{TSS} \cdot \text{ESS}}\rho$.

Step 7: Mathematical implications.

1. Upper and lower bounds on R^2 for a given s . Using $-1 \leq \rho \leq 1$,

$$1 - (1 + s)^2 \leq R^2 \leq 1 - (1 - s)^2. \quad (42)$$

In every case $R^2 \leq 1$. There is no lower bound, because $1 - (1 + s)^2 \rightarrow -\infty$ as $s \rightarrow \infty$.

2. A condition for $R^2 > 0$. The upper bound in implication 1 is not positive when $s = 0$ or $s \geq 2$. So $R^2 > 0$ requires $0 < s < 2$: the vertical spread of the predictions must be less than twice the vertical spread of the actual outcomes. If the predicted spread is more than twice the actual spread, then $R^2 < 0$ with certainty, whatever the direction of the predictions.

3. R^2 never exceeds the squared correlation. Because $(s - \rho)^2 \geq 0$,

$$R^2 = 2\rho s - s^2 = \rho^2 - (s - \rho)^2 \leq \rho^2 \leq 1. \quad (43)$$

Equality, $R^2 = \rho^2$, holds if and only if $s = \rho$, which is the same as $C = 0$. For a fixed direction of the predictions (a fixed ρ), R^2 is largest when the length of the predictions is chosen to be $s = \rho$.

4. The special case of ordinary least squares. With an intercept, fitted and scored on the same data, $C = 0$. Then $s = \rho$ by Step 5 (for $s > 0$), and the following results hold:

$$R^2 = \rho^2 = s^2 = \frac{\text{ESS}}{\text{TSS}}, \quad (44)$$

$$s = \rho \leq 1 \implies \text{ESS} \leq \text{TSS}, \quad (45)$$

$$\text{RSS} = \text{TSS} - \text{ESS} \leq \text{TSS}, \quad (46)$$

$$R^2 \geq 0. \quad (47)$$

So the condition $C = 0$ and the Cauchy–Schwarz inequality together produce the range $0 \leq R^2 \leq 1$ and the equality of R^2 with the squared correlation.

5. Predictions of the form $\hat{y}_i = \bar{y} + k(y_i - \bar{y})$. Then $\rho = 1$ for $k > 0$ and $s = |k|$, so $R^2 = 2k - k^2$ for $k \geq 0$. The maximum of R^2 is 1 at $k = 1$, and the cross-term is $C = k(1 - k)$ TSS.

7 When R^2 is less than -1, the meaning of R^2 changes

The cross-term is often not zero in the examples above re simple linear regression is not fit with idealized OLS, non-linear regression, and out-of-sample fitting. In these cases, R^2 loses some of its meaning because it no longer means the proportion of explained variation, though it can still have other meanings, e.g. in the context of comparisons, it still means how much smaller the model's squared error is compared to the baseline. Another example is below.

R^2 versus vertical compression-stretch.

Setup. There are n actual outcomes y_i , with average $\bar{y}$. Define the predictions by scaling the deviations of the actual outcomes from their average by a factor k :

$$\hat{y}_i = \bar{y} + k(y_i - \bar{y}). \quad (48)$$

The factor k changes only the vertical spread of the predictions. The average of the predictions stays equal to $\bar{y}$, and the x -values are not changed.

$k = 0$	every prediction equals the average (flat)
$0 < k < 1$	predictions are vertically compressed towards the average
$k = 1$	predictions are exact
$k > 1$	predictions are vertically stretched away from the average
$k < 0$	predictions are reversed (on the wrong side of the average)

Step 1: The errors.

$$e_i = y_i - \hat{y}_i = (1 - k)(y_i - \bar{y}). \quad (49)$$

Step 2: The residual sum of squares.

$$\text{RSS} = \sum_{i=1}^n e_i^2 = (1 - k)^2 \sum_{i=1}^n (y_i - \bar{y})^2 = (1 - k)^2 \text{TSS}. \quad (50)$$

Step 3: R^2 as a function of k .

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - (1 - k)^2 = 2k - k^2. \quad (51)$$

This is a parabola that opens downwards, with its maximum at $k = 1$.

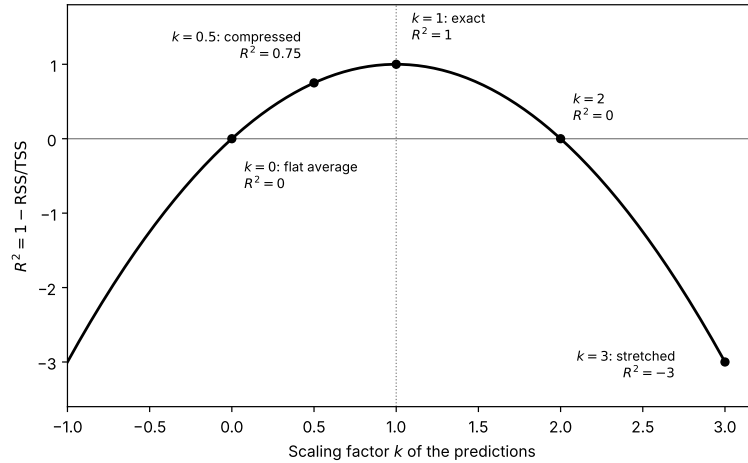


Figure 1: $R^2 = 2k - k^2$ against the scaling factor k .

Step 4: Reading the figure.

k	Meaning	R^2
0	flat average	0
0.5	compressed	0.75
1	exact	1
2	stretched	0
3	stretched	-3
-1	reversed	-3

- R^2 has its maximum value of 1 at $k = 1$, where the errors are zero.
- The curve is symmetric about $k = 1$. Compressing by $k = 0.5$ and stretching by $k = 1.5$ both give $R^2 = 0.75$, because $(1 - k)^2 = 0.25$ in both cases.
- $R^2 = 0$ at $k = 0$ and at $k = 2$. At $k = 0$ the prediction equals the average. At $k = 2$ the prediction lies on the other side of the actual value, at the same distance as the average, so the squared error equals TSS.
- For $k > 2$ or $k < 0$, $RSS > TSS$, so $R^2 < 0$.
- Compression cannot lower R^2 below 0, because the most compressed prediction ($k = 0$) is the flat average. Stretching has no limit, so $R^2 \rightarrow -\infty$ as $|k| \rightarrow \infty$. This is the reason $R^2 = 1 - RSS/TSS$ has an upper bound of 1 and no lower bound.

Step 5: The cross-term on the same axis. Using $TSS = RSS + ESS + 2C$, with $C = \sum_i e_i(\hat{y}_i - \bar{y})$:

$$C = \sum_{i=1}^n (1 - k)(y_i - \bar{y}) \cdot k(y_i - \bar{y}) = k(1 - k) TSS. \quad (52)$$

- $C > 0$ for $0 < k < 1$ (compression), with maximum $C = TSS/4$ at $k = 0.5$.
- $C = 0$ at $k = 0$ and at $k = 1$.
- $C < 0$ for $k > 1$ and for $k < 0$ (stretching and reversal).

The sign of the cross-term does not decide the sign of R^2 . For example, at $k = 1.5$ the cross-term is negative, but $R^2 = 0.75$ is positive.

Step 6: Link to the squared correlation. For every $k > 0$, the predictions are a positive multiple of the actual deviations, so the squared correlation between y and $\hat{y}$ equals 1. At $k = 3$ it is still 1, but $R^2 = -3$. The squared correlation measures only direction. R^2 also measures the size of the errors.